Features: Useful Clues

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Keep the clues the task needs.

Define a size task, expose missing distinctions, compare shelf boxes and reject an answer added after inspection.

Keep one habit: State the task → preserve the distinction → check when the information exists.

MAIN LESSON · SECTION 1

Which clues does this decision actually need?

> Choose task-relevant information that exists when the answer is required.

Start with indistinguishable inputs

Two cards can share a length but need different labels because their widths differ. Before asking for a cleverer model, check whether the input record contains the distinction the task needs.

Retrieve the label-policy check

Two annotators agree that a calendar card is ready. What must you still check before treating that label as correct?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the **relevant** lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can choose recorded properties that preserve distinctions needed for a stated task.
- I can show why identical **feature record**s can hide examples with different **target label**s.
- I can reject an answer copied into the **input**s and test a proposed **feature** with a new case.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

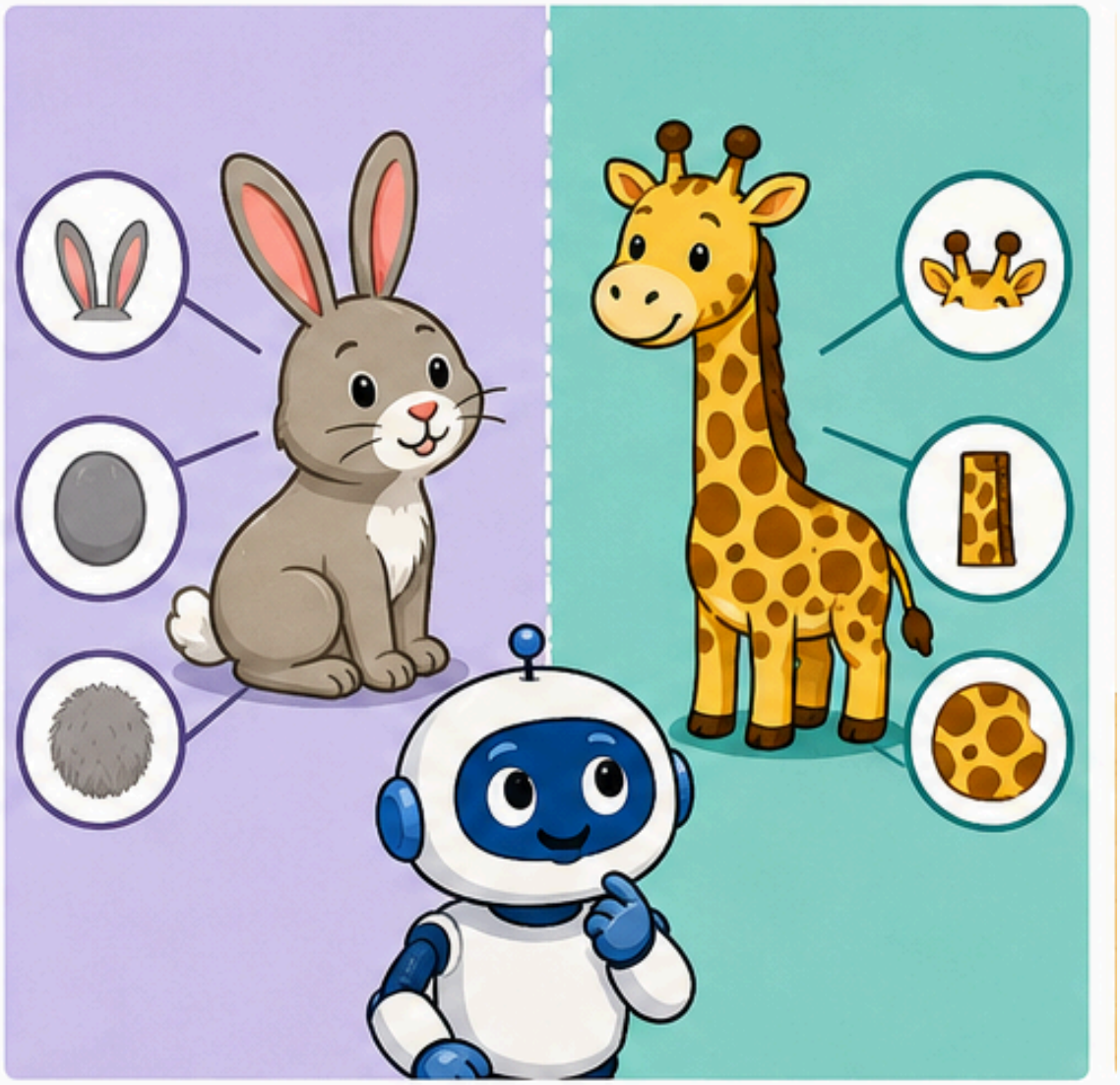
Remember before reading

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Input**, **Process**, **Output**. Give an example and a mistake that the idea helps you avoid.

Some observable clues help a decision more than others.
</details>



LEARNING TARGET A dataset contains only small smooth leaves and large toothed leaves. A model separates them by size. Design two fresh examples that distinguish a size shortcut from an edge-based rule.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can choose task-**relevant feature**s and avoid a clue created after the decision.

MAIN LESSON · SECTION 2

Define the task before choosing features

> The delivery task uses two inclusive size limits under a declared paper convention.

Read the full card task

Imagine a paper sorting task for a model delivery station. Each rectangular card has a length, width, colour and card number.

The station labels a card fits if its length is at most 12 centimetres and its width is at most 8 centimetres. Both limits include equality.

Cards stay flat in the marked orientation; turning, bending and thickness are outside this paper task. All measurements in the first examples are known positive whole numbers.

Separate features, identifiers and the target

A **feature** is a recorded **input** property, such as width in centimetres. The **target label** is the answer we want to produce: fits or too large under this guide. Decide what the answer means before choosing the **input**s.

A card number can help us find a record, but knowing its number does not tell us its dimensions.

Recognise when a direct rule is sufficient

This guide already gives us a direct rule, so we do not need to train a model to solve it. We use the task to investigate the information that either a rule or a **learned model** would need.

Later, when relationships are learned from examples, the same question remains: did we supply the distinctions the task requires?

Inspect the candidate inputs

- State the label rule and the fixed orientation. Which recorded properties decide the label? Which property only identifies a card, and what information would not yet exist before inspection?

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY Ear shape, muzzle length, and paw shape may help distinguish cats from dogs; background color and photo border usually do not.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or **distraction**.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

MAIN LESSON · SECTION 3

The same feature record can hide different answers

> Keep the necessary dimensions, their order and their units.

Read the two cards

Card A is 10 centimetres long and 7 centimetres wide. Card B is 10 centimetres long and 9 centimetres wide. A fits; B is too large because its width exceeds 8. Compare two ways of recording these cards.

Compare short and fuller records

The same cards under two feature choices

ORIGINAL CARD	CHOOSE THE INPUTS	RESULTING FEATURE RECORD
A is 10 by 7 cm and fits	Keep length only	10 cm
B is 10 by 9 cm and is too large	Keep length only	10 cm
A is 10 by 7 cm and fits	Keep length and width	10 cm and 7 cm
B is 10 by 9 cm and is too large	Keep length and width	10 cm and 9 cm

Length alone makes A and B look identical to the system. Retaining width preserves the difference needed by this task. The letters identify the examples here; they are not additional inputs.

Explain what the missing width prevents

A system receiving only the length cannot use the hidden width to tell these two cards apart. The same **input**, 10, is paired with two different correct answers.

It may guess, return an uncertainty, or use a usual answer, but length alone cannot guarantee the right label for both. A more complicated model cannot reconstruct this missing distinction with certainty from that **input** alone.

State the feature convention

A **feature record** needs a consistent order and units. Here we write length in centimetres first, then width in centimetres.

The record 10, 7 means something different if someone silently switches the order or records millimetres. Write the definitions alongside the data.

Explain a deciding arrow

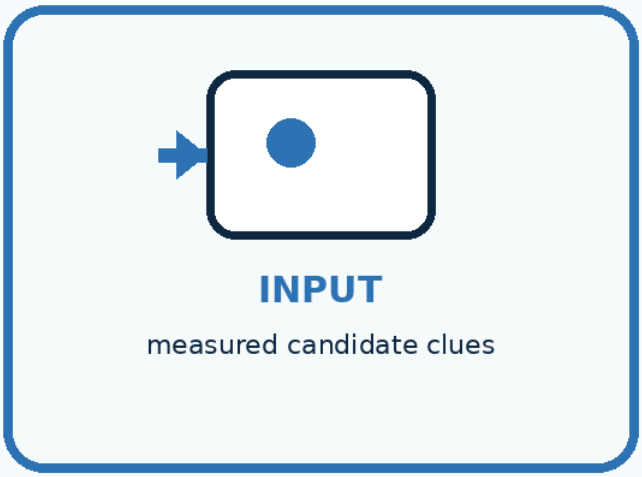
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

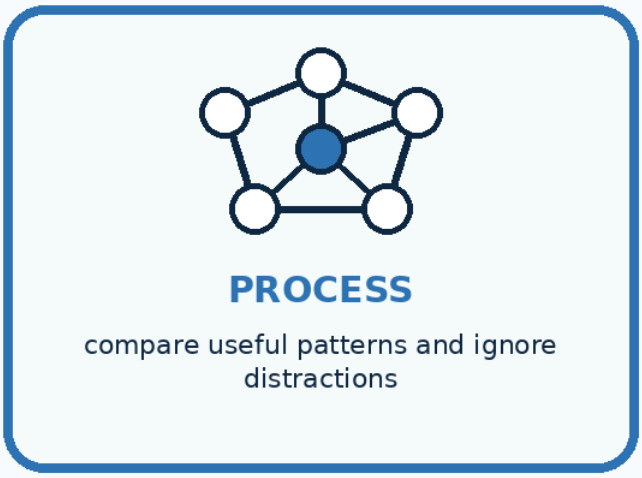
▼ Earlier wording, alternatives and picture transcripts

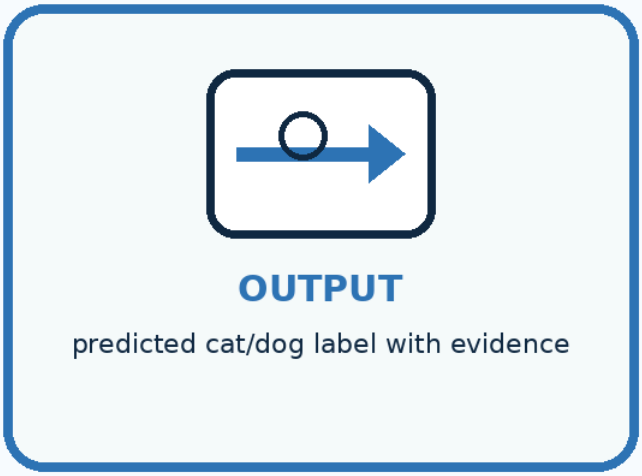
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT measured candidate clues

PROCESS compare useful patterns and ignore **distraction**s

OUTPUT predicted cat/dog label with **evidence**

CORE IDEA Some observable clues help a decision more than others.

UNDER THE HOOD A **feature** is a represented property used by a decision **process**.
Predictive usefulness depends on the task and data.

Spurious **feature**s correlate in a sample without reflecting the intended concept and often fail when conditions change.

Draw a fresh example with all three stages:

A closer explanation

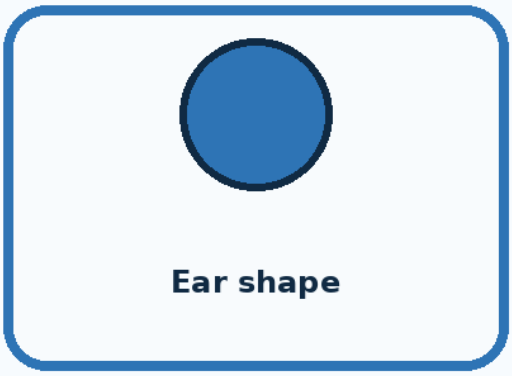
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare useful clues and tempting shortcuts

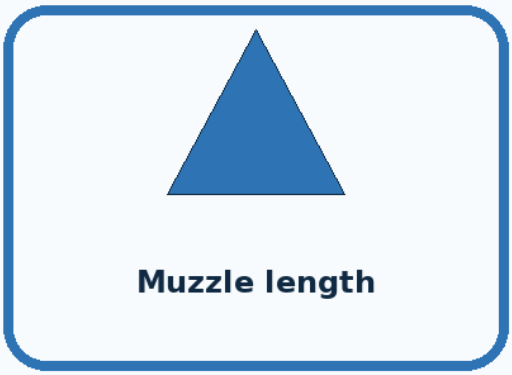
> Ask what the stated task needs and whether a clue could fail on a deliberately different case.

Pictures paired with their evidence



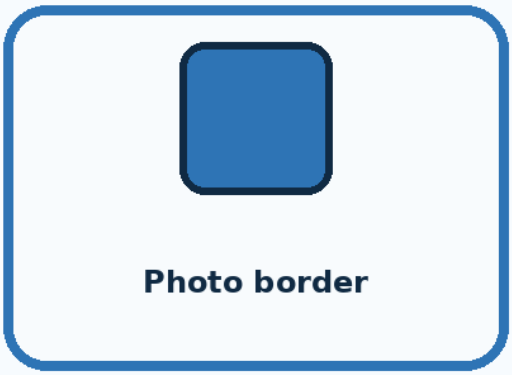
The case to investigate

CASE 1 — Ear shape Clue: may differ across many examples. Meaning: potentially useful.



The case to investigate

CASE 2 — Muzzle length Clue: observable animal trait. Meaning: potentially useful.



The case to investigate

CASE 3 — Photo border Clue: added by the photographer. Meaning: **distraction**.



The case to investigate

CASE 4 — Grass background Clue: accidental setting pattern. Meaning: **spurious clue**.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

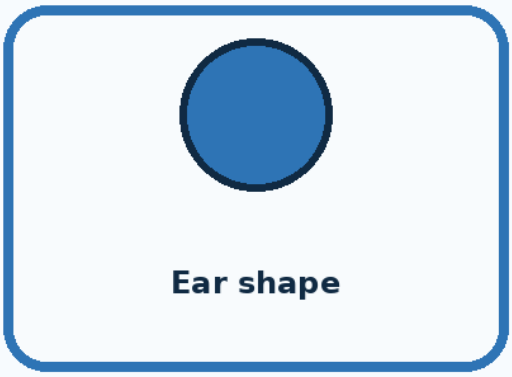
BOUNDARY A clue can be observable yet still be ir**relevant**, unfair, or fragile.

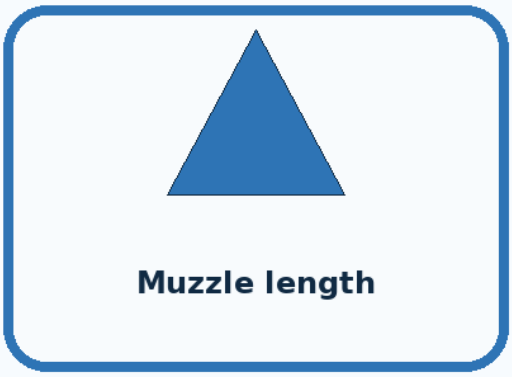
▼ Earlier wording, alternatives and picture transcripts

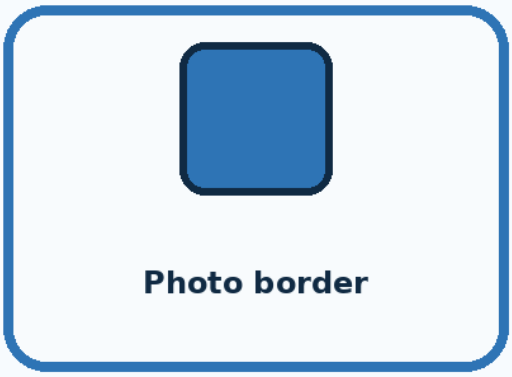
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Ear shape Clue: may differ across many examples. Meaning: potentially useful.

CASE 2 — Muzzle length Clue: observable animal trait. Meaning: potentially useful.

CASE 3 — Photo border Clue: added by the photographer. Meaning: **distraction**.

CASE 4 — Grass background Clue: accidental setting pattern. Meaning: **spurious clue**.

Change one thing in the pictured case

Change: Change Q's width to 6 without changing its height.

Prediction: Q fits the shelf.

Reason: Both dimensions now meet their limits, with equality allowed; stripe pattern does not change the rule.

MAIN LESSON · SECTION 6

Dimensions decide this shelf task

> Keep the two needed measurements and reject a sticker added after the decision.

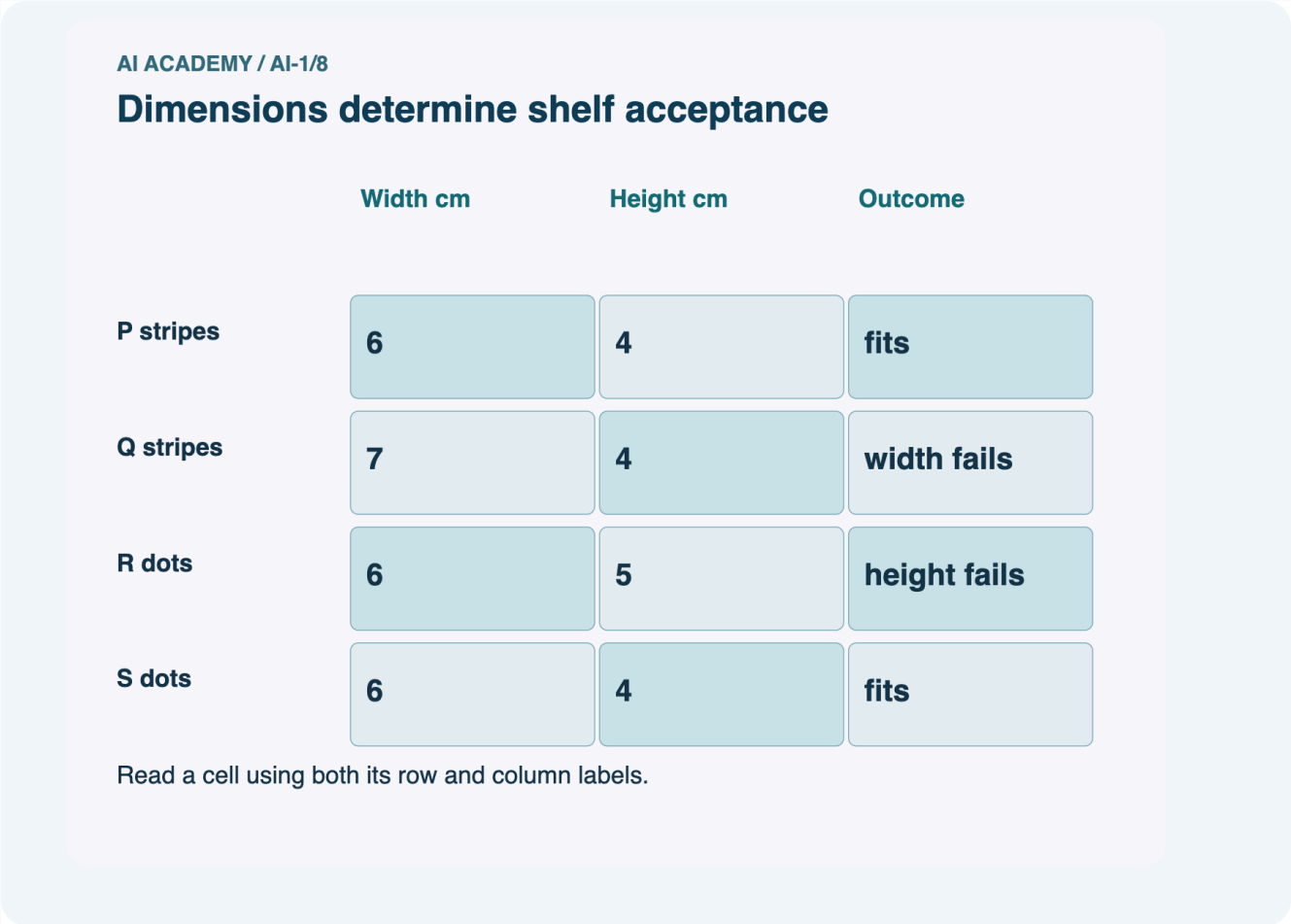
Shelf rule, box records and later sticker

A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.

Boxes P=(6,4,stripes), Q=(7,4,stripes), R=(6,5,dots), S=(6,4,dots).

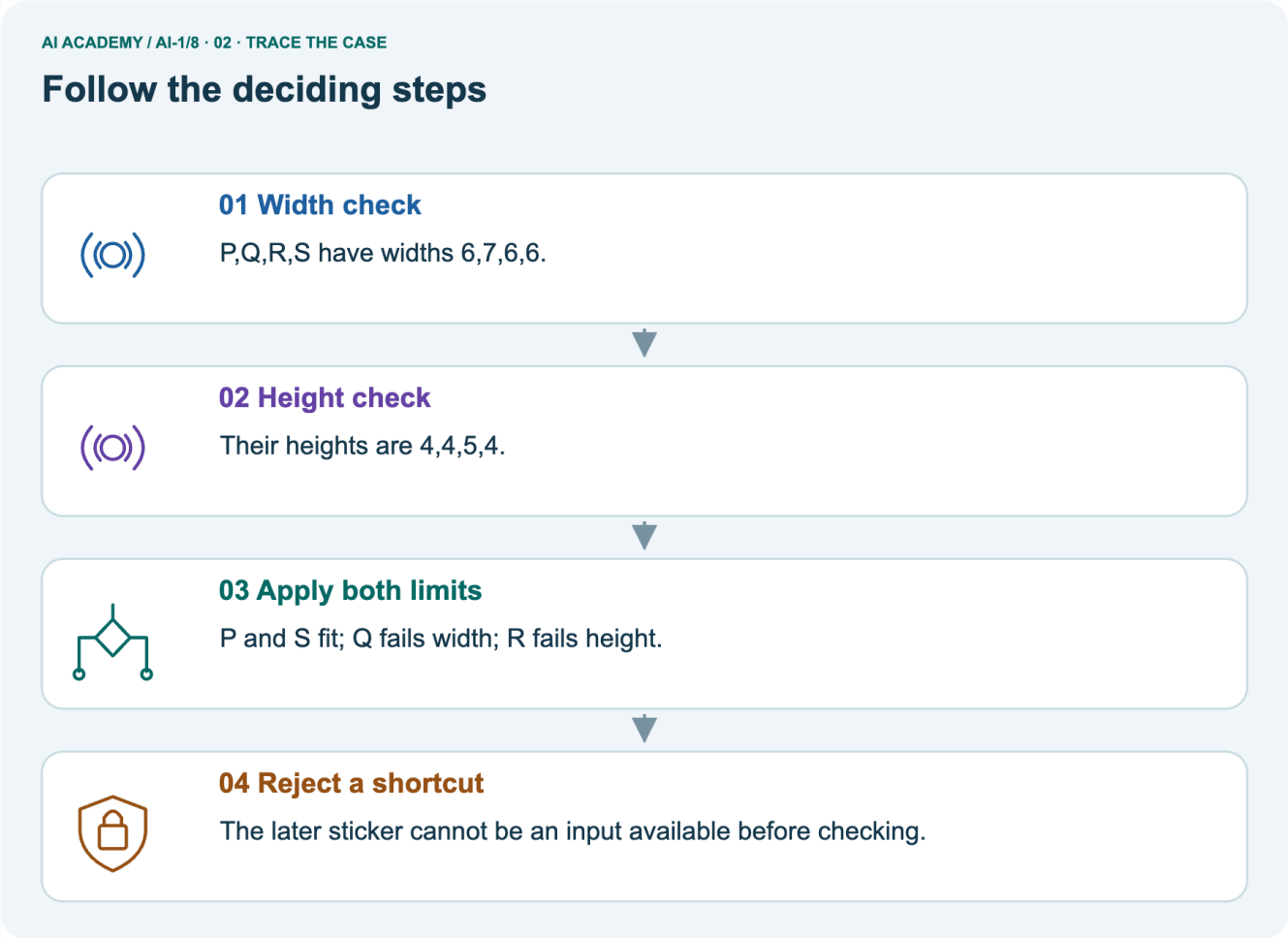
A green acceptance sticker is attached only after the shelf check.

Read the compared dimensions



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace both checks and the information timing



Follow the labelled picture with the matching explanation. Open the vector diagram

Use pairs that isolate one change

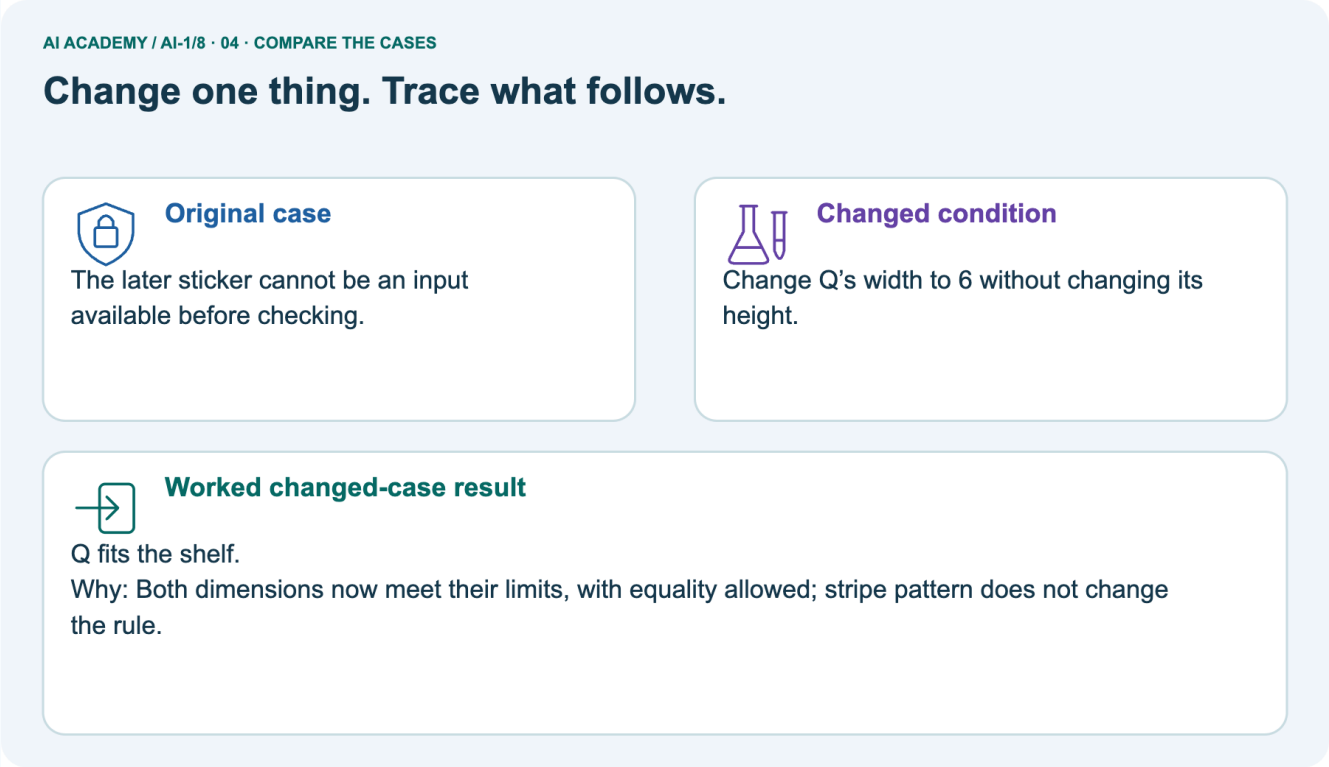
Comparing P with Q isolates width; comparing S with R isolates height. Comparing P with S changes pattern while both remain accepted. Under this declared policy, pattern is not a deciding **feature**.

A later acceptance sticker leaks the very answer the system should produce.

Predict Q at the width boundary

- Before checking, change only Q’s width from 7 to 6 cm and retain its height of 4 cm. Apply both inclusive limits. Which other box now shares its two dimensions, and does the stripe pattern decide the answer?

Check the changed dimensions



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit the claim to this task

These comparisons establish the written shelf rule, not the importance of pattern for unrelated tasks.

Explain the pattern comparison and leakage

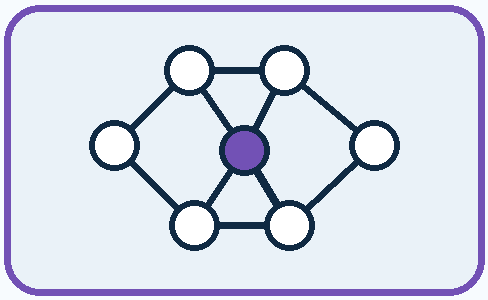
- Read the labelled picture: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE Ear shape, muzzle length, and paw shape may help distinguish cats from dogs; background color and photo border usually do not.

STEP 1 — OBSERVE measured candidate clues

STEP 2 — TRACE compare useful patterns and ignore **distraction**s

STEP 3 — CONCLUDE predicted cat/dog label with **evidence**

MODEL REASONING Useful **feature**s describe the animal and remain available across settings. Borders and backgrounds describe the photo setup.

Even ear shape may fail for unusual breeds or blocked views, so multiple clues and uncertainty are needed.

DO NOT OVERCLAIM A clue can be observable yet still be ir**relevant**, unfair, or fragile.

What new **evidence** could change this answer?

A closer explanation

Card measurements and colour	Label under the guide	What the comparison shows
10 by 7 cm and blue	Fits	Compare with the next card
10 by 9 cm and blue	Too large	Same length and colour but width changes the label
13 by 7 cm and blue	Too large	Same width and colour as the first card but length changes the label
10 by 7 cm and red	Fits	Changing only colour does not change this guide

The first two cards show why width matters. The first and third show why length matters. The first and fourth show that colour has no part in this particular rule.

That is a statement about our task, not a claim that colour is useless for every task. For sorting red and blue cards, colour would be essential.

Suppose every **fitting** card in a small practice set is blue and every oversized card is red. Colour would appear to predict the labels perfectly in that set. A red 10 by 7 card and a blue 13 by 7 card expose the shortcut.

Check a proposed **feature** on deliberately different cases before trusting the practice score.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Features: Useful Clues

The facts we will use

AI ACADEMY / AI-1/8

Dimensions determine shelf acceptance

	Width cm	Height cm	Outcome
P stripes	6	4	fits
Q stripes	7	4	width fails
R dots	6	5	height fails
S dots	6	4	fits

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/8

Features: Useful Clues

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Width check

P,Q,R,S have widths 6,7,6,6.

2. Height check

Their heights are 4,4,5,4.

3. Apply both limits

P and S fit; Q fails width; R fails height.

4. Reject a shortcut

The later sticker cannot be an input available before checking.

CHANGE ONE THING

Change Q's width to 6 without changing its height.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Width check	P,Q,R,S have widths 6,7,6,6.
2. Height check	Their heights are 4,4,5,4.
3. Apply both limits	P and S fit; Q fails width; R fails height.
4. Reject a shortcut	The later sticker cannot be an input available before checking.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Change Q's width to 6 without changing its height. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Test the representation before trusting a practice score

> Identical inputs can conceal different targets, and a changed task may need different inputs.

Use a two-answer counterexample

Before adopting a **feature** set, state the task, name the **input**s with units, and ask whether two different answers can share the same **input** record. Construct such a pair if you can.

Then check that each **input** is available at the time the answer is needed. These checks can reveal a broken representation before any model is trained.

Respect the paper-task assumptions

For our known measurements and fixed orientation, length and width are sufficient to apply the paper rule exactly.

That does not establish whether a real parcel will fit through a real opening: thickness, shape and measurement error were excluded. If the task changes, review the **feature**s again.

More columns are not automatically better; each one needs a reason to be collected and used.

Repair and explain the colour shortcut

- CLAIM A Use every available clue because **more information** is always better.
- CLAIM B A **feature** that worked once will always work.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

> **Someone claims that a model using only card colour is reliable because it matches every practice label. Give a new card that challenges this claim and explain what should be checked instead**

▼ Earlier wording, alternatives and picture transcripts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Ear shape	may differ across many examples	potentially useful
Muzzle length	observable animal trait	potentially useful
Photo border	added by the photographer	distraction
Grass background	accidental setting pattern	spurious clue

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Draw the line at decision time

> The answer sticker is useful for teaching or checking, but it is unavailable for this earlier decision.

Find the leaked target

After measuring a card, a teacher adds a sticker saying fits or too large. A learner proposes using sticker text as an **input** to predict the label before that inspection.

The sticker seems excellent in the completed records because it contains the answer. But it does not exist at the time of the intended **prediction**. This is **leakage**: development has used information the real task cannot provide.

Draw a line at the moment the system must answer. On the before side, put the permitted measurements available then. On the after side, put the final label sticker.

Keep labels available for teaching and checking **prediction**s, but do not feed the sticker back in as an **input** for this task.

Explain the incomplete pre-decision record

> **A card is 12 cm long and its width is unreadable. Someone proposes using the label sticker that will be added after inspection. Which information is usable before inspection? Explain why it does not establish the label**

Use W2 to challenge a one-feature record

Use workbook W2 for the connected example "**Feature**s and useful distinctions". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge “a stronger model restores every omitted value”

- Use the worked case for **Feature**s and useful distinctions. Proposed explanation: “A stronger model can always recover an omitted measurement”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the pictured sticker

Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.

> **My explanation using a pictured step and the given facts**
>
> ___
>
> ___
>
> ___
>
> ___

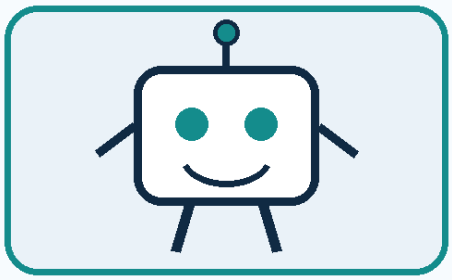
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Some observable clues help a decision more than others.

Use a fresh example to explain **feature**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: A **feature** that worked once will always work.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Prove that one measurement is insufficient

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate: test candidate clues on contrasting cards

> Use supplied records to challenge a shortcut and limit claims beyond the observed examples.

Your fictional picture records

Use four fictional animal-picture records. A: cat, pointed ears, short muzzle, curved tail, blue background. B: dog, floppy ears, long muzzle, straight tail, red background. C: cat, pointed ears, short muzzle, straight tail, red background. D: dog, floppy ears, long muzzle, curved tail, blue background. These are supplied toy observations, not biological rules for all cats and dogs.

Predict, compare and challenge a shortcut

1. Record three candidate visual clues: ear shape, muzzle length and tail shape. Before inspecting all four labels, predict which might help. Compare the observations with the supplied labels, then test a background-only guess on C and D. Explain which associations hold in these four records and what remains untested. For a new record with unreadable ear and muzzle details, name the missing evidence instead of using its background to claim a certain species.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep observations and limits separate

Case	Prediction	Observed result	What it means
Normal			
Edge			

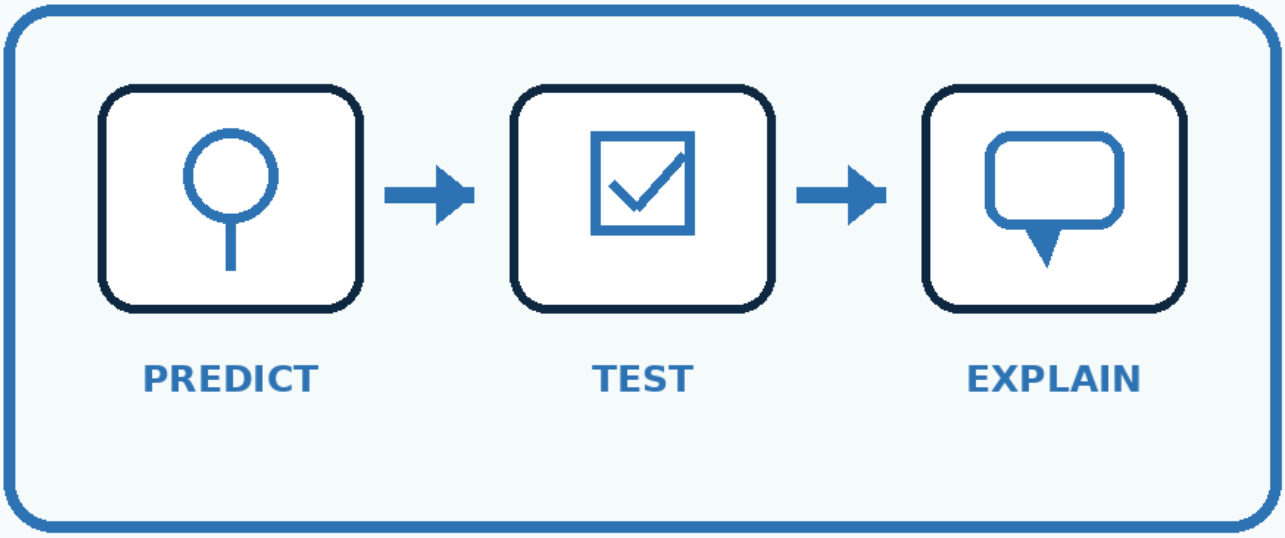
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EXPLORER LAB

Clue Keep-or-Reject

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Choose three clues for cat/dog sorting; reject **distraction**s.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: design a record for a different tray

> Use the new height and thickness limits, including equality, and keep the later sticker out of the inputs.

The new task

A fictional display tray accepts a tile if its height is at most 9 cm and its thickness is at most 2 cm, including equality. Other dimensions are fixed. Before checking, you can record height, thickness and pattern name.

A pass sticker is added afterward. Choose **input**s with units. Construct two tiles with the same height but different labels.

Then label a 9 cm high, 2 cm thick striped tile and explain why the sticker is unsuitable as an **input** before checking.

> **My independent explanation, trace and limits**

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Design a record for a new task

Week 8 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A sorter separates round and square tiles. Both shapes come in red and blue. Choose a useful **feature** and a distracting **feature**. Test your choice on a blue circle.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one **relevant** detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Select the smallest useful **feature**s for lost-and-found sorting and reject clues tied to a particular classroom or owner.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A dataset contains only small smooth leaves and large toothed leaves. A model separates them by size. Design two fresh examples that distinguish a size shortcut from an edge-based rule.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep only justified pre-decision features

> Connect each collected property to the sorter’s stated task.

This week’s design decision

THIS WEEK'S CHECKPOINT Select the smallest useful **feature**s for lost-and-found sorting and reject clues tied to a particular classroom or owner.

Choose sorter **feature**s that exist at decision time and actually connect to the stated classroom requirement.

Plan the features, checks and review

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Keep relevance and fragility in view

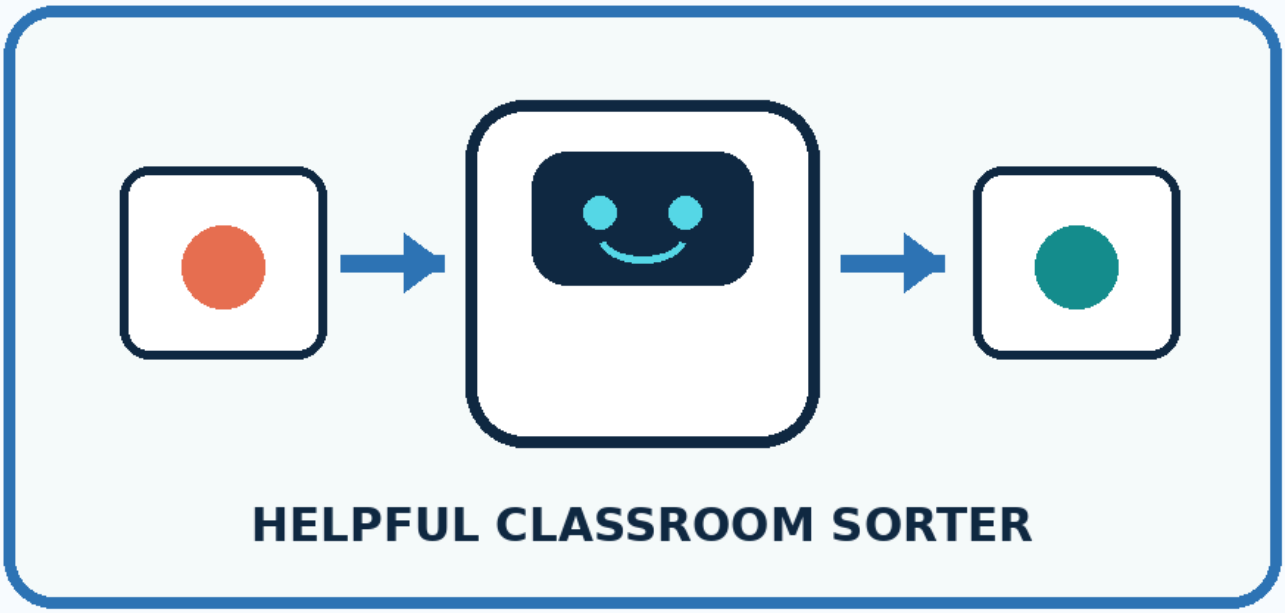
RESPONSIBLE DESIGN A clue can be observable yet still be ir**relevant**, unfair, or fragile.

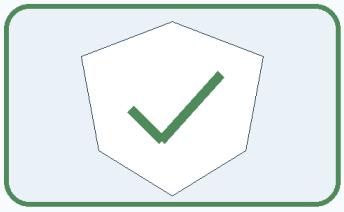
▼ Earlier wording, alternatives and picture transcripts

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: defend a feature and reject a shortcut

> Close your notes and explain why a particular distinction must be retained.

Your exit explanation

- Close your notes. Explain this goal in your own words: Choose recorded properties that preserve distinctions needed for a stated task

My deciding step, evidence and one remaining question

Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
feature	represented clue used in a decision	
relevant	connected to the task goal	
distraction	available clue that should not drive the decision	
spurious clue	accidental pattern that may fail later	

Terms for the connected explanation

Word	Meaning
Target label	The answer a supervised learning task asks a model to predict
Feature record	The selected input values for one example, in a defined order

Word	Meaning
Relevant feature	An input property that helps make distinctions needed for the stated task
Leakage	Information used during development that would not legitimately be available when making the intended prediction

Repair a claim using a word

- CLAIM TO REPAIR Use every available clue because **more information** is always better.
- Use one vocabulary word and one piece of **evidence** to repair it:

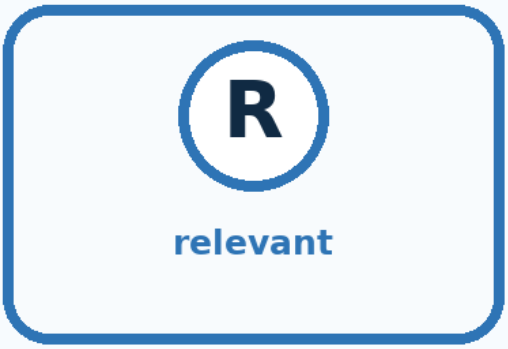
▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

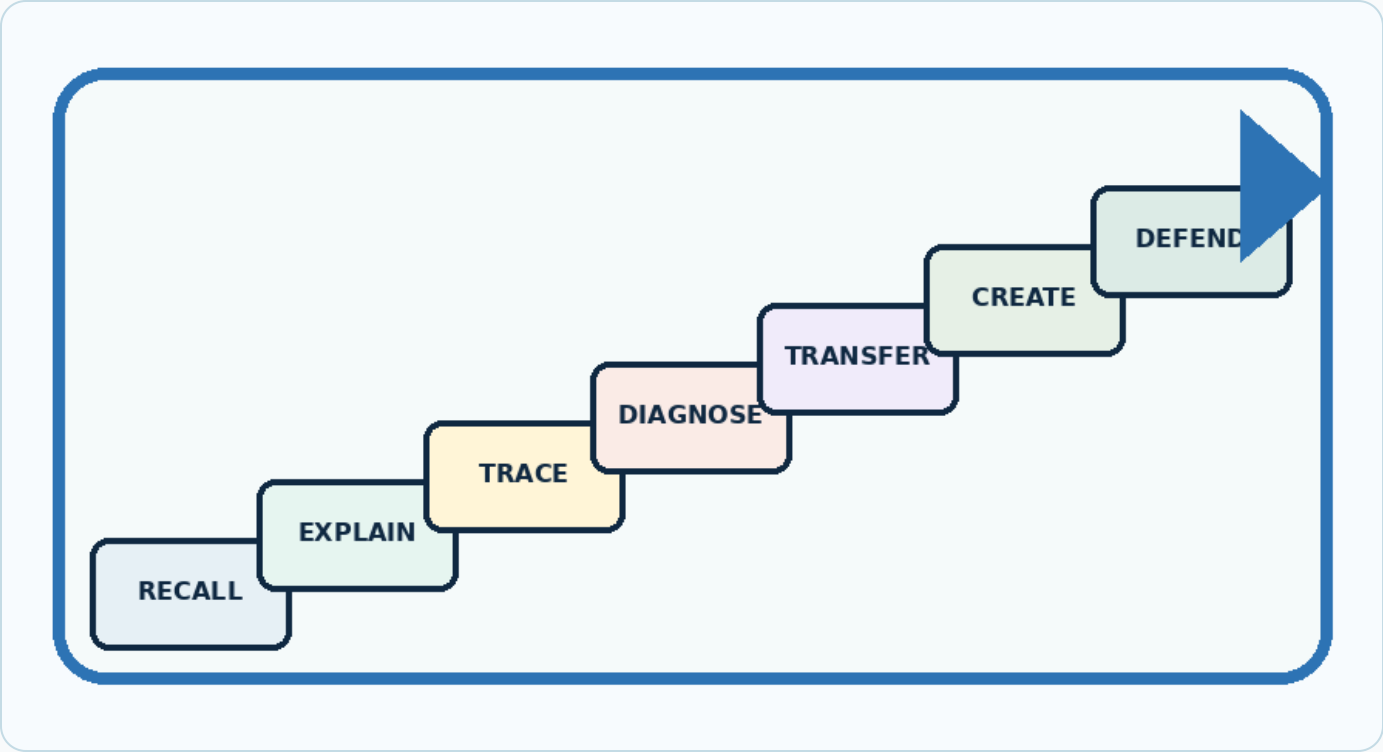
- RECALL Define **feature**.
- EXPLAIN Some observable clues help a decision more than others.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: Use every available clue because **more information** is always better.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Design a record for a new task is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

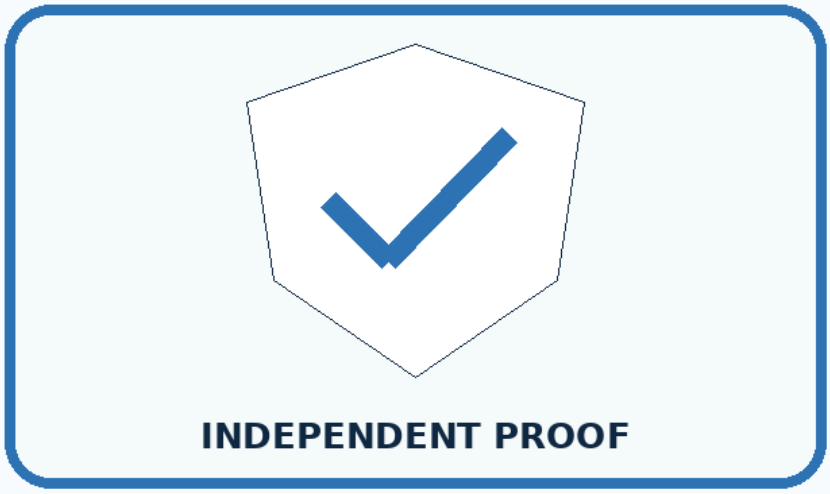
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Design a record for a new task is the selected independent case.



EXPLANATION CHECK State the task and units, retain distinctions needed by the rule, test a counterexample and reject information unavailable at decision time.

Independent **evidence**:

SELF-CHECK Useful **feature**s describe the animal and remain available across settings. Borders and backgrounds describe the photo setup.

Even ear shape may fail for unusual breeds or blocked views, so multiple clues and uncertainty are needed.

NEXT BRIDGE Carry today's **evidence** habit into: Algorithms as Recipes.

▼ **Extra practice: Choose a clue that answers this task**

REFERENCE · SECTION 13

Extra practice: Choose a clue that answers this task

> Try the question before opening its worked explanation.

Read the supplied case and question

Two green leaves have smooth and toothed edges. The task is to distinguish edge types.

Choose the **relevant feature** and explain why size is insufficient.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Edge shape directly addresses the target. Size can vary within both categories. **Feature** usefulness depends on the task.

Evidence or stage	Value or result
Target	Smooth versus toothed edge
Useful clue	Edge shape
Weak clue	Leaf size alone

▼ Earlier wording, alternatives and picture transcripts

Week 8 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Choose a clue that answers this task

Two green leaves have smooth and toothed edges. The task is to distinguish edge types.

Choose the **relevant feature** and explain why size is insufficient.

Edge shape directly addresses the target. Size can vary within both categories. **Feature** usefulness depends on the task.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Choose recorded properties that preserve distinctions needed for a stated task	Explain the deciding step, show a trace or test, and state one limit.
Show why identical feature records can hide examples with different target labels	Explain the deciding step, show a trace or test, and state one limit.
Reject an answer copied into the inputs and test a proposed feature with a new case	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.
- Which labelled step matches this detail? The later sticker cannot be an **input** available before checking.
- 1. Apply both limits
- 2. Reject a shortcut
- 3. Height check
- 4. Width check
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported explanation: Reject a shortcut The later sticker cannot be an **input** available before checking. Comparing P with Q isolates width; comparing S with R isolates height.

Comparing P with S changes pattern while both remain accepted. Under this declared policy, pattern is not a deciding **feature**. A later acceptance sticker leaks the very answer the system should produce.

Changed-case question

- Changed-case practice: Change Q's width to 6 without changing its height. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. Q fits the shelf.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: Q fits the shelf. Both dimensions now meet their limits, with equality allowed; stripe pattern does not change the rule.

Repair and save your work

Review Categories and Labels (ai-1/7). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Explain why B originally was revise rather than review, then label its changed version..

Return to this week and explain what you changed.

Review Pictures Become Numbers (ai-1/5). Goal: Encode and reconstruct a small image using an explicit grid convention.

Short practice: Find a position where the original badges differ, then read the changed A back as a picture.. Return to this week and explain what you changed.

Review **Input**, **Process**, **Output** (ai-1/2). Goal: Trace information through a system without confusing information with a physical action. Short practice: Use the richer record to find blue pencils in Box B.

Explain why total alone was insufficient.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

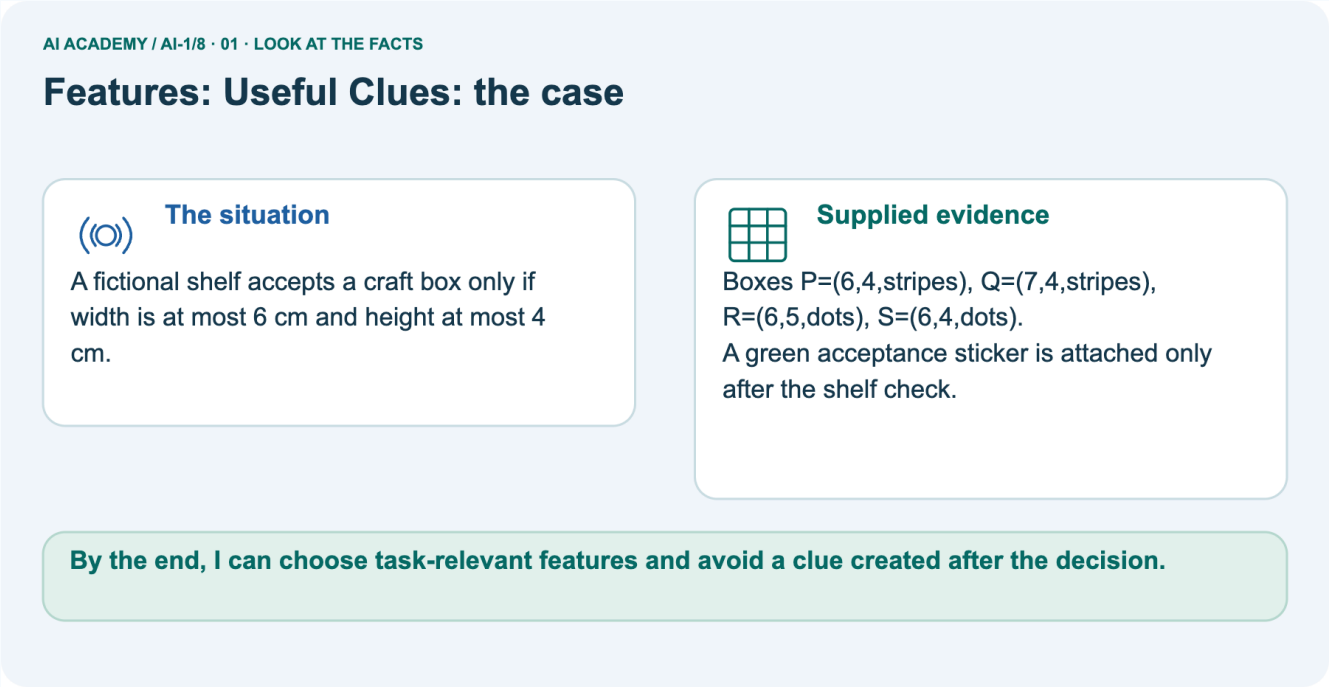
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Features: Useful Clues: the case



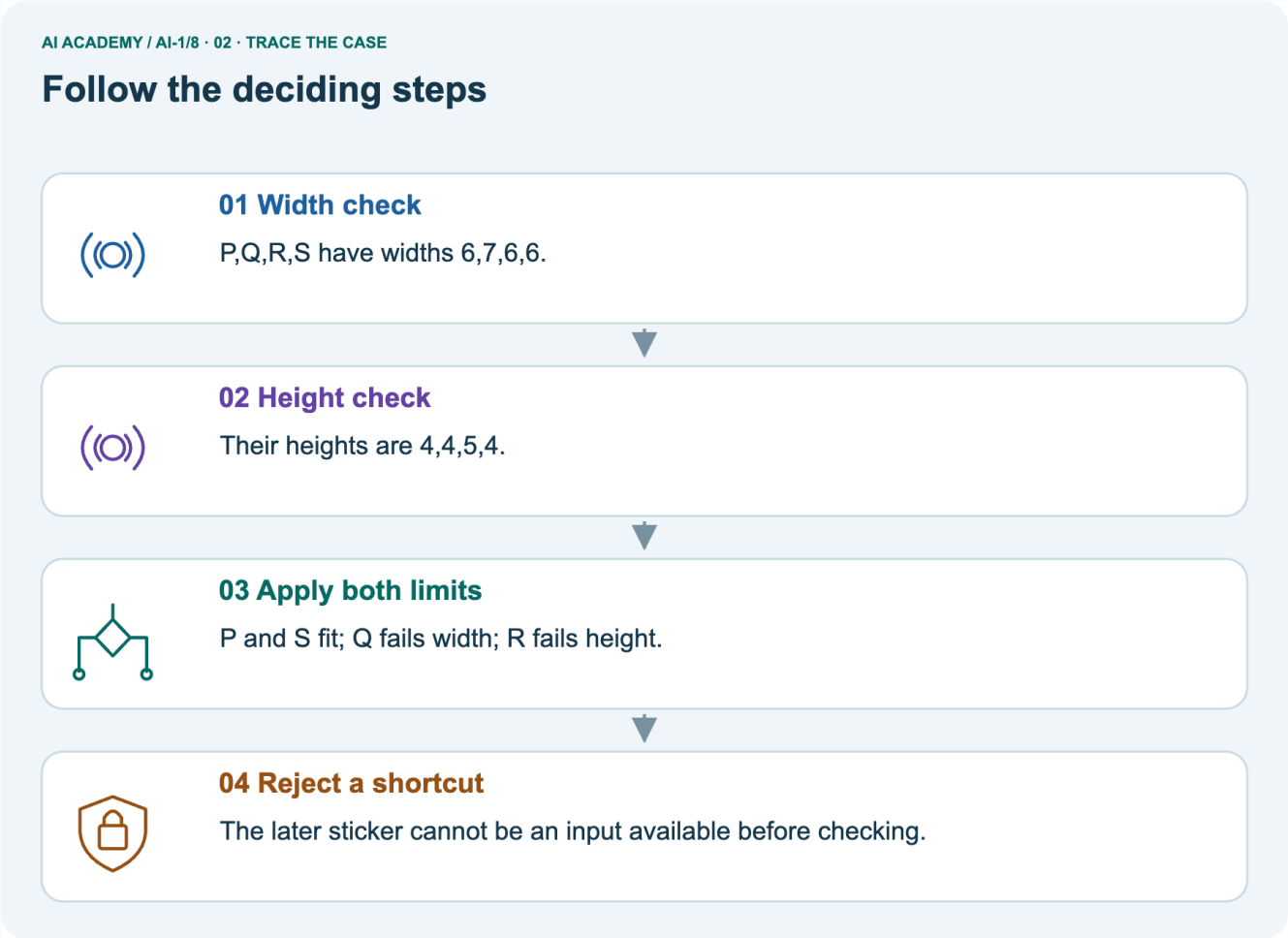
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.
Supplied facts: Boxes P=(6,4,stripes), Q=(7,4,stripes), R=(6,5,dots), S=(6,4,dots).

A green acceptance sticker is attached only after the shelf check.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

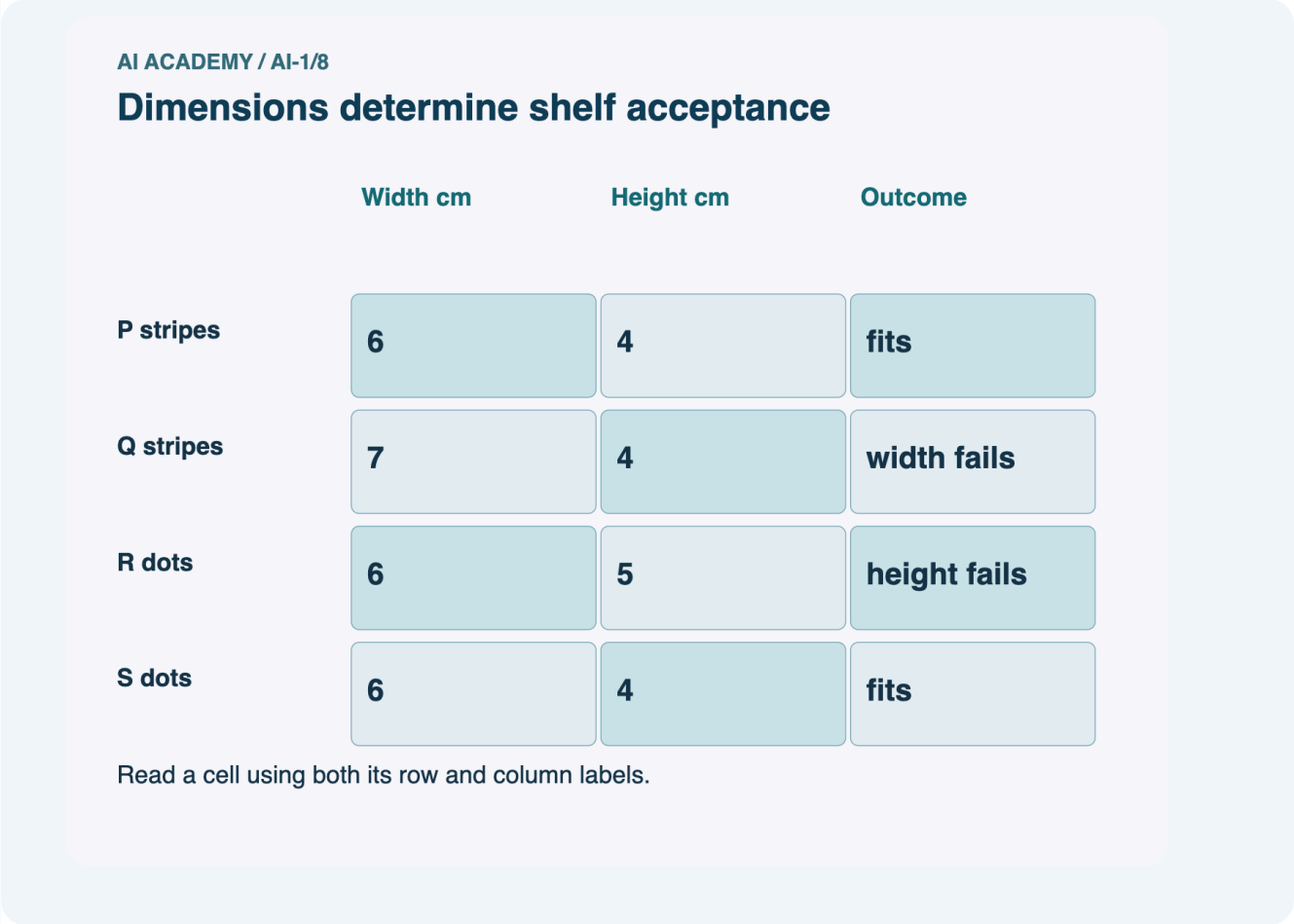


Follow the labelled picture with the matching explanation. Open the vector diagram

Width check: P,Q,R,S have widths 6,7,6,6. → Height check: Their heights are 4,4,5,4. → Apply both limits: P and S fit; Q fails width; R fails height. → Reject a shortcut: The later sticker cannot be an **input** available before checking.

Point to each numbered stage and explain which recorded **input** it uses.

03. Dimensions determine shelf acceptance



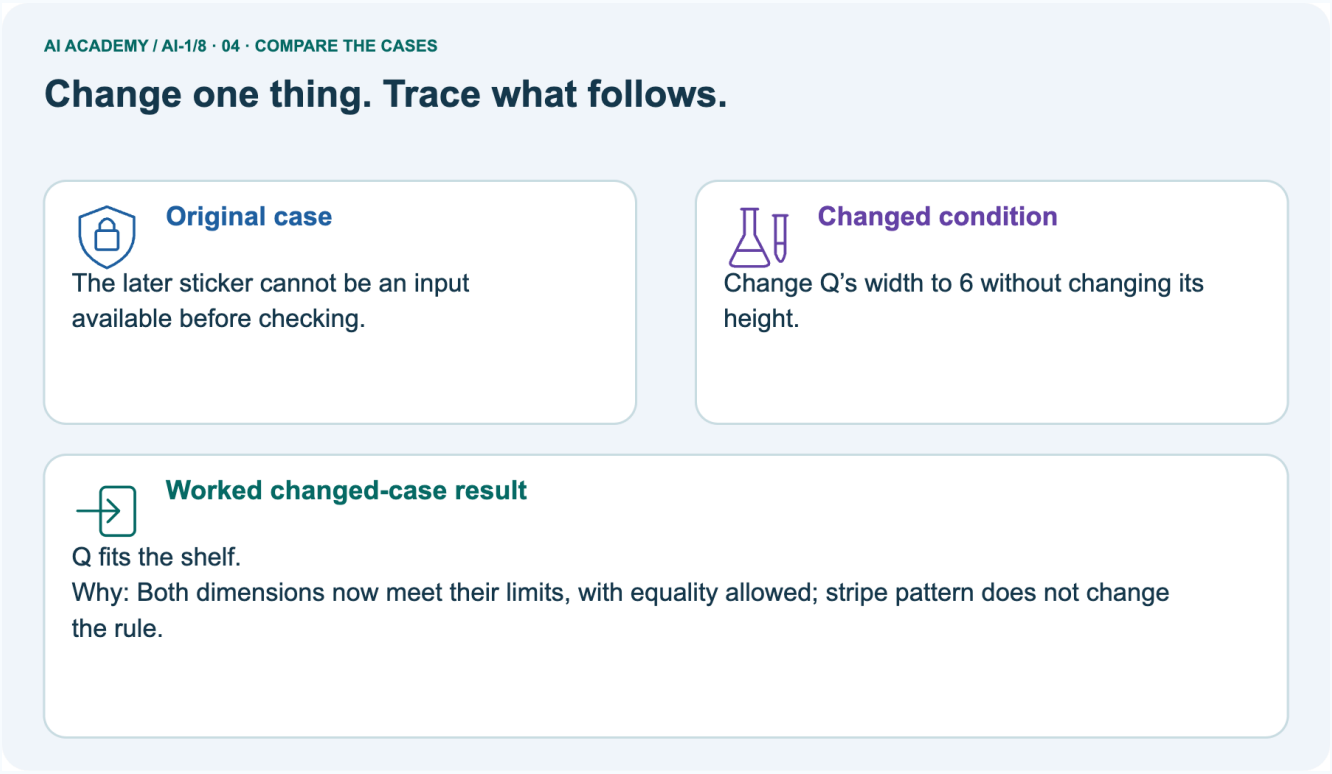
Follow the labelled picture with the matching explanation. Open the vector diagram

Dimensions determine shelf acceptance. P stripes: Width cm = 6, Height cm = 4, Outcome = fits. Q stripes: Width cm = 7, Height cm = 4, Outcome = width fails. R dots: Width cm = 6, Height cm = 5, Outcome = height fails.

S dots: Width cm = 6, Height cm = 4, Outcome = fits.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



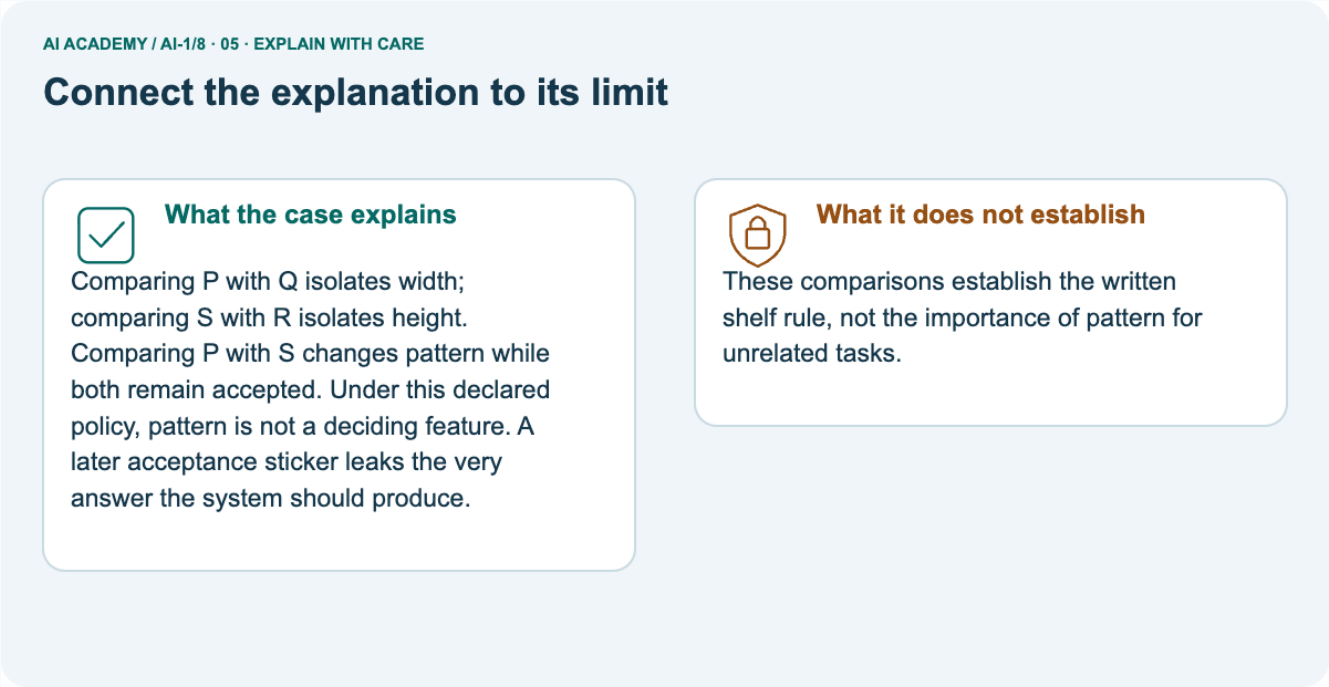
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The later sticker cannot be an **input** available before checking. Changed condition: Change Q's width to 6 without changing its height. Expected result: Q fits the shelf.

Reason: Both dimensions now meet their limits, with equality allowed; stripe pattern does not change the rule.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



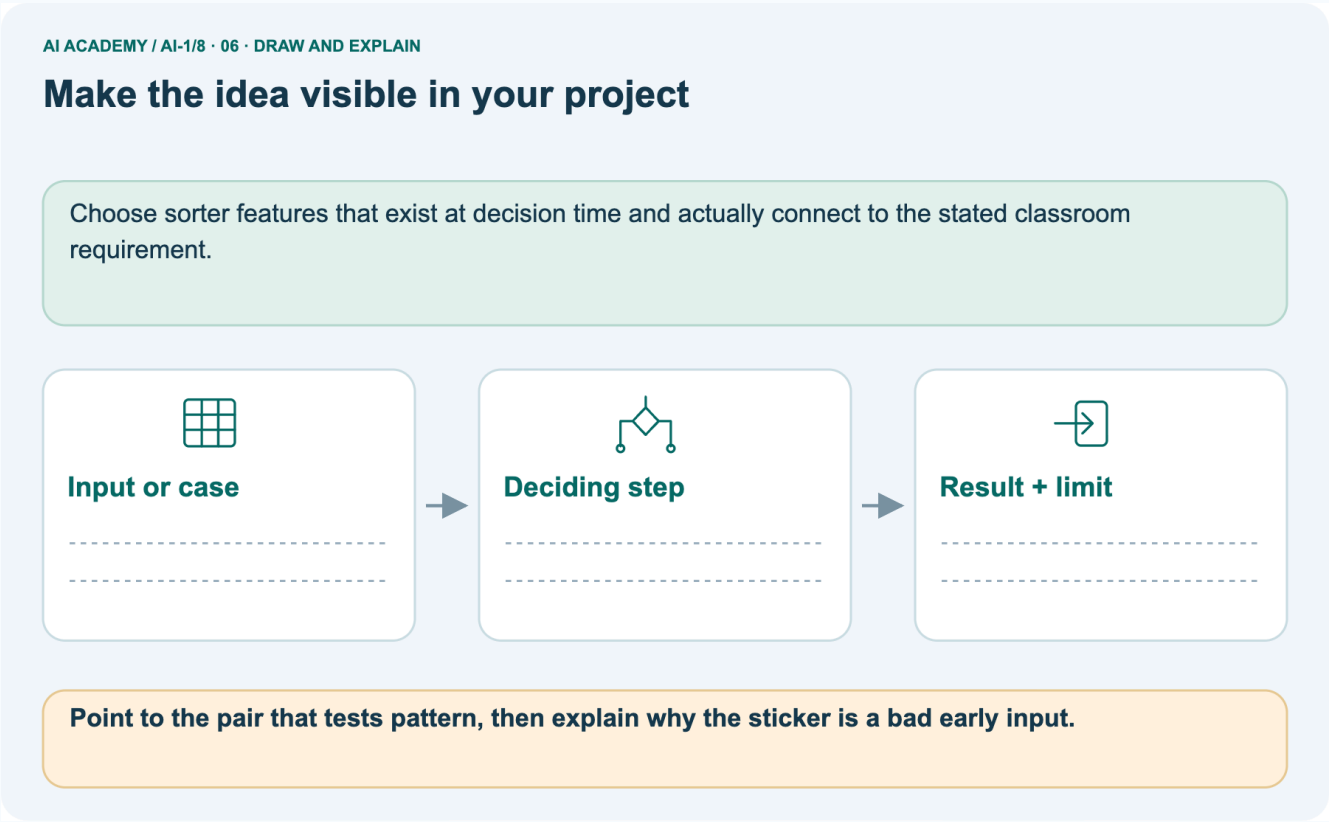
Follow the labelled picture with the matching explanation. Open the vector diagram

Comparing P with Q isolates width; comparing S with R isolates height. Comparing P with S changes pattern while both remain accepted. Under this declared policy, pattern is not a deciding **feature**.

A later acceptance sticker leaks the very answer the system should produce. Limit: These comparisons establish the written shelf rule, not the importance of pattern for unrelated tasks.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose sorter **feature**s that exist at decision time and actually connect to the stated classroom requirement. Draw your **input**, deciding step, result and limit.

Supported question: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Features: Useful Clues: the case

02. Follow the deciding steps

03. Dimensions determine shelf acceptance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.